Title: A Comparative Study of Quantum LDPC Codes for Quantum Chemistry Simulation

Year: 2023

Authors: A. Ahmed, L. Haddad, J. Kowalski, K. Lindqvist (Institute of Data and Intelligent Systems, Summit College)

Keywords: decoherence, Hamiltonian simulation, NISQ, quantum gates, variational algorithms

Abstract:

Fault-tolerant logical gate synthesis remains a critical challenge for Hamiltonian simulation, particularly in quantum chemistry simulation. This paper proposes a framework built on quantum LDPC codes that explicitly accounts for qubit decoherence suppression. We evaluate the framework on simulated Hamiltonians of H2 and LiH molecules and compare it against 5 baseline methods. Experimental results show that the proposed approach improves gate fidelity by 20.7% while keeping circuit success probability within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of Hamiltonian simulation has changed how researchers and practitioners approach fault-tolerant logical gate synthesis, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, quantum LDPC codes have attracted sustained attention as a promising route toward solving fault-tolerant logical gate synthesis in realistic settings such as quantum chemistry simulation. Despite considerable progress, existing solutions for fault-tolerant logical gate synthesis tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in approximation ratio frequently come at the expense of gate fidelity, forcing designers to accept trade-offs that are poorly understood.

Our work is motivated by the growing demand for quantum chemistry simulation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of fault-tolerant logical gate synthesis, a practical design built on quantum LDPC codes, and an empirical evaluation on simulated Hamiltonians of H2 and LiH molecules. Understanding qubit decoherence suppression is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Larsen et al. (2008) introduced an early framework based on quantum LDPC codes, reporting a 10.0% improvement in gate fidelity over previous baselines. Subsequent work by Nakamura et al. (2019) extended this line of research to quantum chemistry simulation, although the evaluation was restricted to small-scale settings. A survey by OConnor et al. (2021) categorised existing approaches to fault-tolerant logical gate synthesis into three families and highlighted the lack of standard benchmarks. Müller et al. (2009) analysed qubit decoherence suppression and showed that purely heuristic solutions cannot guarantee consistent circuit success probability. More recently, Lindqvist et al. (2014) combined quantum LDPC codes with trapped-ion gate architectures, achieving a 23.2% reduction in overhead on simulated Hamiltonians of H2 and LiH molecules.

Several studies have criticised the reliance on synthetic workloads, arguing that results on simulated Hamiltonians of H2 and LiH molecules may not generalise to quantum chemistry simulation. In contrast to these efforts, our work treats fault-tolerant logical gate synthesis and qubit decoherence suppression jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to

fault-tolerant logical gate synthesis as a persistent obstacle for quantum chemistry simulation. Despite this body of literature, no prior study has systematically compared quantum LDPC codes and trapped-ion gate architectures under identical experimental conditions.

3. Methodology

We formalise fault-tolerant logical gate synthesis as a constrained optimisation task in which the objective is to maximise gate fidelity while keeping approximation ratio within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on quantum LDPC codes, and a validation layer that incorporates trapped-ion gate architectures. All experiments were conducted on simulated Hamiltonians of H2 and LiH molecules, which provides 128 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 20 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies quantum LDPC codes to reduce the effect of qubit decoherence suppression, followed by a calibration step that stabilises approximation ratio. We compare against 5 baseline methods, including a conventional approach built on trapped-ion gate architectures and two recent variants reported in the literature. The proposed design improves gate fidelity by 20.7% relative to the strongest baseline and reduces gate fidelity by 20.2%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the trapped-ion gate architectures component reduces gate fidelity by approximately 6.4%, confirming its contribution. A sensitivity analysis over 500 parameter settings shows that performance remains stable, with variation in approximation ratio below 7.9%. Threats to validity include the limited diversity of simulated Hamiltonians of H2 and LiH molecules and the possibility that deployment in quantum chemistry simulation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated fault-tolerant logical gate synthesis through the lens of quantum LDPC codes, and demonstrated that a carefully designed pipeline can improve gate fidelity by 20.7% on simulated Hamiltonians of H2 and LiH molecules. The findings suggest that Hamiltonian simulation will continue to benefit from closer integration of quantum LDPC codes and trapped-ion gate architectures, particularly in quantum chemistry simulation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of qubit decoherence suppression. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for fault-tolerant logical gate synthesis.